

Searching 2: Binary Search Problems

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Reachable on Scaler Lounge!

“The art of searching is knowing what to look for,
where to look and how to look for it”

Q. Given a sorted integer array where every element appears twice except for one element. Find that unique element.

$A = [2, 2, 5, 5, 8, 10, 10, 13, 13, 18, 18]$

Solⁿ (1) $\rightarrow$ Ans = $\bigoplus^{\text{ans}} A[i]$ TC = $O(N)$, SC = $O(1)$

Solⁿ (2) $\rightarrow$ linear search
 $\forall i$ check if $A[i] \neq A[i-1]$ and $A[i] \neq A[i+1]$ TC = $O(N)$, SC = $O(1)$

Solⁿ (3) $\rightarrow$ Using hashmap TC = $O(N)$, SC = $O(N)$

⊗ 4 pillars of problem solving

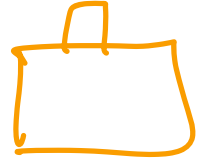
(1) Understanding the problem

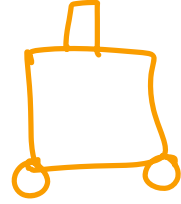
(2) Coming up with brute force solutions

(3) Comparing the solⁿs / coding the solution

(4) Testing our solution (Dry running our code)

* wheel  $\Rightarrow$ < 10000 years back

② trolley  $\Rightarrow$ < 5000

$\Rightarrow$ ③  $\longrightarrow$ 1970 $\leftarrow$ 50 years back

① Understand the problem.

	Array	Ans	
<u>ex ①</u> *	$A = []$	↓	
	$A = [1]$	null	*
	$A = [1, 2]$	1	
		1, 2	
$\longrightarrow$	$A = [a]$	a	
	$[1, 2]$	$\times$	
$\rightarrow$ *	$A = [1, \underline{1}]$	$\times$	

$\nwarrow$ edge

if $N == 1$
return $A[0]$

obsⁿ

$A.size() = 2$

X

→ X $A = [1\ 1\ 1]$

X

→ $A = [1\ 1\ 2]$ 2

→ $A = [10\ 10\ 20\ 20\ 30]$ 30

$A = [10\ 10\ 15\ 20\ 20]$ 15

ex. $A = [2\ 2\ 5\ 5\ 4\ 8\ 10\ 10\ 13\ 13\ 18\ 18]$

Diagram illustrating the array structure with indices 0 to 10. The array is divided into two groups: the first group (indices 0-4) contains 5 elements, and the second group (indices 5-10) contains 4 elements. The element 4 is highlighted in a blue box, and the element 8 is highlighted in a blue box. The element 10 is highlighted in a blue box. The element 13 is highlighted in a blue box. The element 18 is highlighted in a blue box.

① If neighbors are not equal to the element
⇒ ans = element

② find the first index of this repeating

element then try to count elements remaining on the left side.

① int findUnique (int A[], int N) {

② l = 0, r = N-1

$L > R$

2a if (N == 1) return a[0];

2b if (a[0] != a[1]) return a[0];

2c if (a[N-1] != a[N-2]) return a[N-1];

③ while (l <= r) { // search space

④ mid = (l+r)/2

// what we are looking for

⑤ if (A[mid] != A[mid-1] and
⑥ A[mid] != A[mid+1])
⑦ return a[mid];

(8)
 (9)
 (10)
 (11)
 (12)
 (13)
 (14) }

if (A[mid] == A[mid-1])
 |
 mid = mid - 1

if (mid % 2 == 0)
 |
 l = mid + 2

else
 |
 r = mid - 1

A = [1]

l $\Rightarrow$ 0
 r $\Rightarrow$ 0
 mid $\Rightarrow$ 0

A = [^l2⁰ ¹2¹ ^m5² ³5³ ^r8⁴ ^m10⁵ ⁶10⁶ ⁷13⁷ ⁸13⁸ ⁹18⁹ ^r18¹⁰]

l $\Rightarrow$ 0 $\rightarrow$ 4
 r $\Rightarrow$ 10 $\rightarrow$ 4
 mid $\Rightarrow$ 8 $\rightarrow$ 2 $\rightarrow$ 4

Ans (8)

10:42

Q.

Given a sorted array, which is rotated.

Find out index of
(1) pivot element

(2) any given element

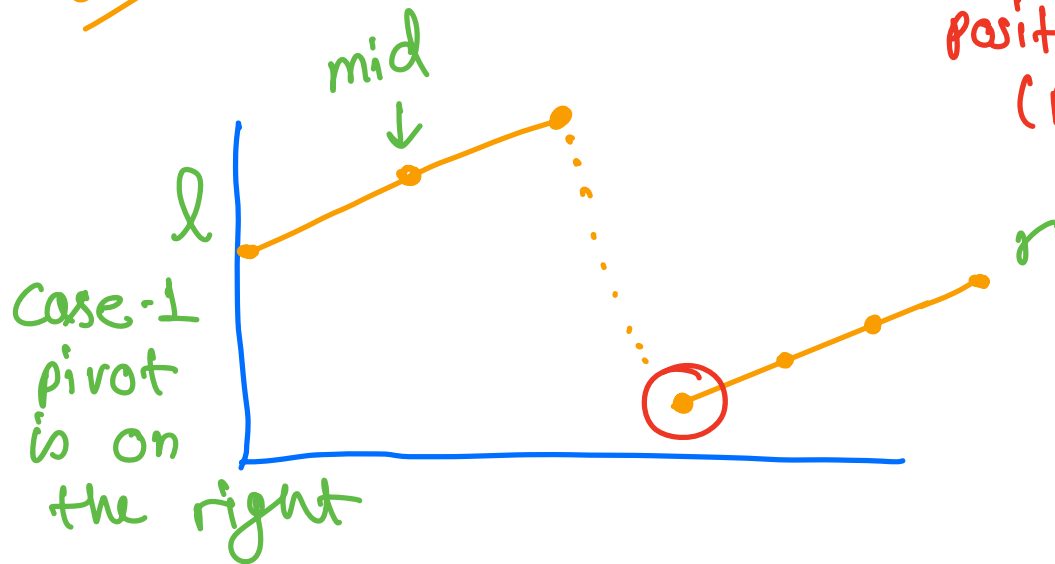
$$A = [1]$$

$$A = [10 \ 20 \ 30 \ 40 \ 50 \ 60 \ 70]$$

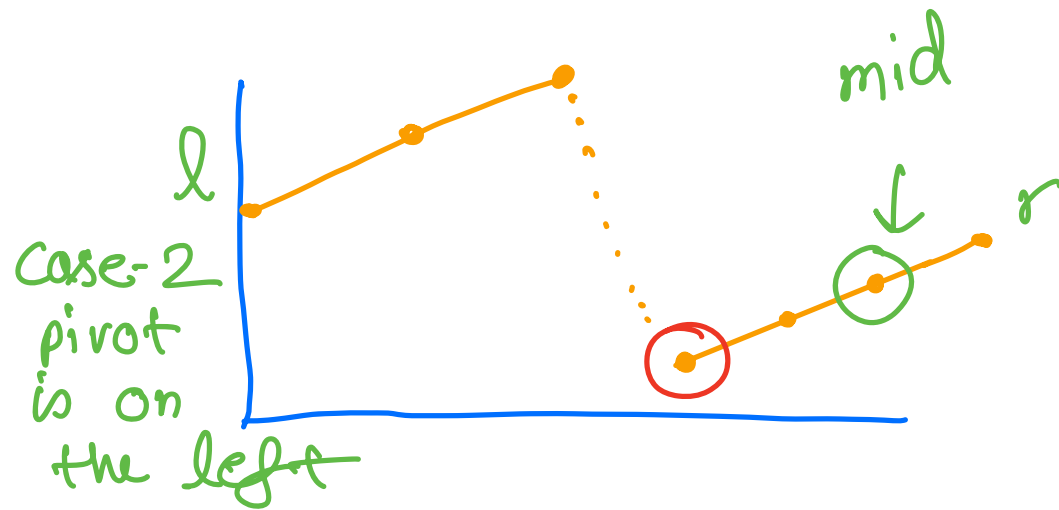
input array

$$A = [50 \ 60 \ 70 \ 10 \ 20 \ 30 \ 40]$$

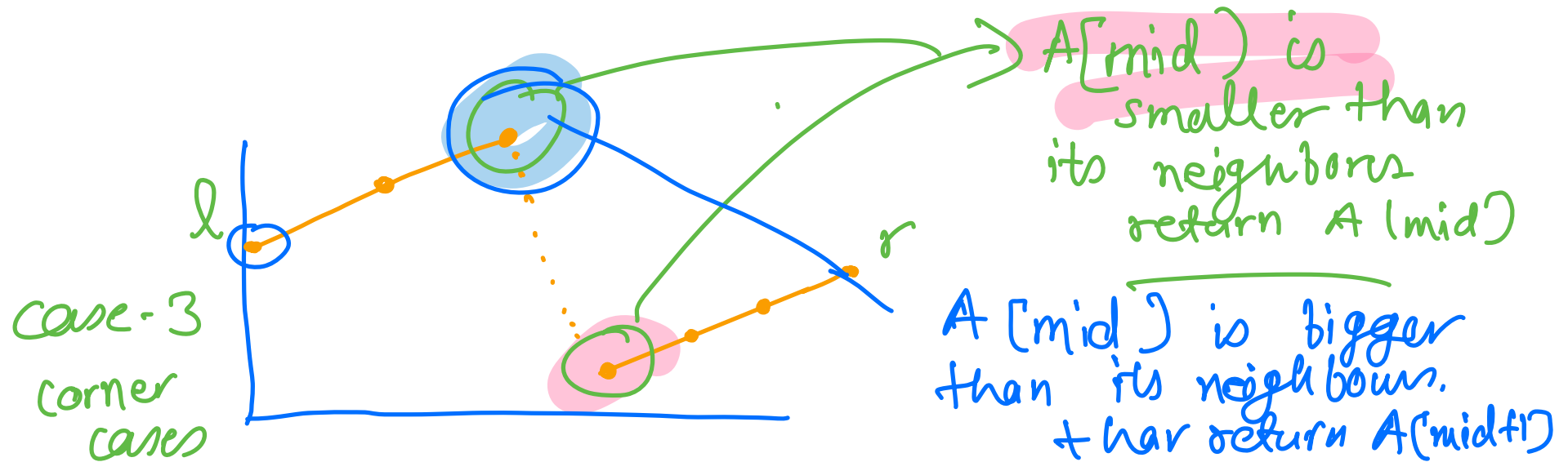
position of the smallest element
(pivot)



If $(A[mid] > A[r])$
move right



if $(A[mid] < A[l])$
more left



recursive code
hw. $\rightarrow$ iterative

```

int findPivot (int[] A, l, r) {
    if (l > r) return -1;
    if (l == r) return l;
    mid = (l + r) / 2;
    if (A[mid] < A[mid-1] and
        A[mid] < A[mid+1])
        return mid;
    if (A[mid] > A[mid-1] and
        A[mid] > A[mid+1])
        return mid+1;
    if (A[l] > A[mid])
        return findPivot (A, l, mid-1)
    else {
        return findPivot (A, mid+1, r)
    }
}

```

A = [10]

3 4

$$A = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 20 & 25 & 35 & 10 \end{pmatrix}$$

$$\begin{aligned} l &\rightarrow 0 \rightarrow 2 \\ r &\rightarrow 3 \\ \text{mid} &\rightarrow \frac{0+3}{2} \rightarrow 1 \rightarrow 2 \end{aligned}$$

$$\boxed{\text{ans} = 3}$$

Q Given a number find the square root of the number without library functions.

(int)
N

i/p
100
110
70
25

o/p
10
10
8
5

$$(\sqrt{70}) = 8.36 \sim 8$$

Brute force $a = \sqrt{N} \rightarrow$ a +ve number where $a * a = N$

(2S)

search space $[1 \rightarrow N]$

$$O(\sqrt{N})$$

70

$$1 \times 1 = N \quad \times$$

$$2 \times 2 = 4 < 25 \quad \times$$

$$3 \times 3 = 9 < 25 \quad \times$$

$$4 \times 4 = 16 < 25$$

$$5 \times 5 = 25 = 25$$

$$6 \times 6 = 36 > 70$$

$$7 \times 7 = 49 < 70$$

$$8 \times 8 = 64 < 70$$

$$9 \times 9 = 81 > 70$$

$$\Rightarrow \sqrt{N} = 5$$

$$\text{ans} = 8$$

[1 ... - ... N]

mid * mid < N
→ right

mid * mid > N
25 ← left

]

25

1	2	3	4	5	6
0	0	0	0	1	1

$i * i < N$

Q U.W find the square root upto 3 decimal digits.

$$N = 10, \quad \sqrt{10} = 3.162 \text{ (ans)}$$

Q



Q